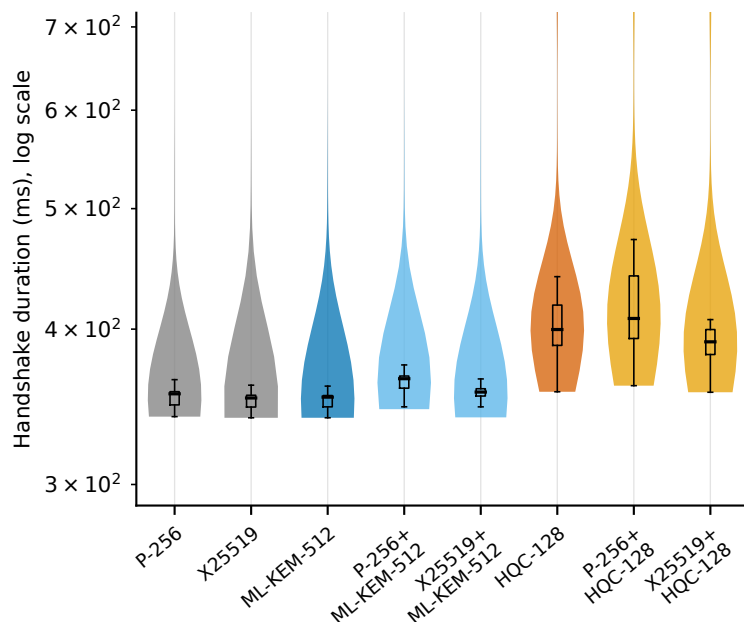
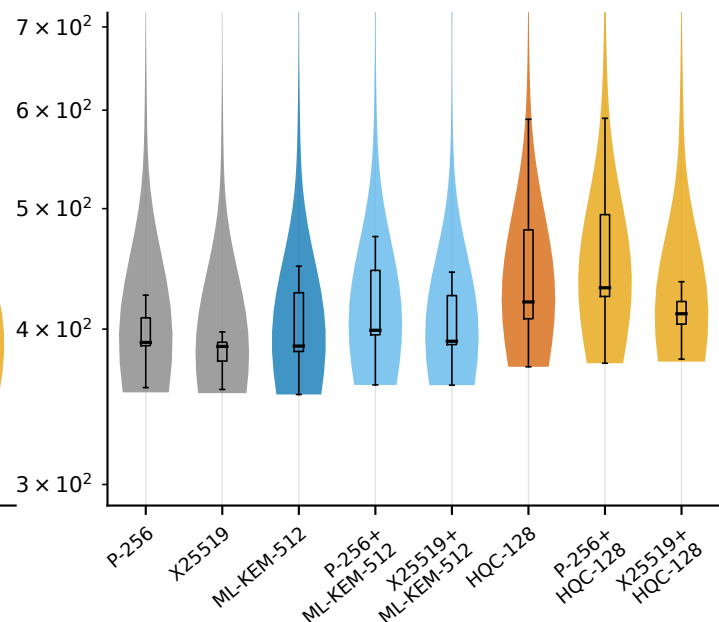


(a)

ed25519 (classical)

**(b)**

mlds44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mlds44 Mean (ms)	mlds44 CV (%)	mlds44 Max (ms)
P-256	379.7	39.5	1.37k	436.6	49.6	2.38k
X25519	382.8	46.2	2.36k	424.1	47.8	1.75k
ML-KEM-512	383.2	44.5	1.55k	435.9	44.2	1.81k
P-256+ML-KEM-512	387.9	42.4	2.06k	452.6	43.5	1.81k
X25519+ML-KEM-512	381.9	41.9	1.56k	440.2	44.3	1.98k
HQC-128	432.6	37.0	1.76k	471.4	43.8	1.99k
P-256+HQC-128	454.5	43.4	1.81k	490.2	42.9	1.98k
X25519+HQC-128	434.1	41.8	1.74k	457.7	47.6	2.25k